

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-7446
Project Name	ComicCrafts – AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3 (Tailwind CSS)	Component-based UI for the story prompt box, Generate button and comic panel display. Responsive design works on both desktop and mobile.
2	Backend / Server-Side	Python with FastAPI	Fast asynchronous handling of AI requests. Python gives simple integration with the Gemini SDK for prompt validation and comic generation.
3	Database / Data Storage	Firebase Firestore (NoSQL) and Firebase Storage	Stores story prompts, comic outlines and panel data as flexible JSON (Prompt and Comic Datastores). Storage keeps generated panel images for later viewing and download.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	Cloud / Hosting / Deployment	Google Cloud Run, Firebase Hosting, Docker	Easy auto-scaling and pay-per-use hosting. Docker gives consistent deployment, and Google Cloud works smoothly with the Gemini API.
5	Version Control & CI/CD	Git, GitHub, GitHub Actions	Seamless team collaboration and code tracking. GitHub Actions automates testing and deployment on every push.
6	Third-Party APIs / Other Tools	Google Gemini API (text and image models), html2canvas / jsPDF	Gemini creates the structured comic outline, panel-wise scenes, image prompts and panel images from the user's story idea. html2canvas / jsPDF let the user save or download the finished comic.